Solution:

$$\begin{aligned} u &= x^2y - x^3 + xy^2 - y^3 \\ \frac{\partial u}{\partial x} &= 2xy - 3x^2 + y^2 \\ \frac{\partial u}{\partial y} &= x^2 + 2xy - 3y^2 \\ \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} &= -2x^2 - 2y^2 + 4xy \\ &= -2\left(x^2 + y^2 - 2xy\right) \\ &= -2\,(x-y)^2 \end{aligned}$$

Example 6.36

If $u = \log(x^2+y^2)$, then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Solution:

$u = \log(x^2+y^2)$

$$\frac{\partial u}{\partial x} = \frac{1}{x^2+y^2}(2x) = \frac{2x}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\left(x^2+y^2\right)\cdot 2 - 2x\cdot 2x}{\left(x^2+y^2\right)^2} = \frac{2\left(y^2-x^2\right)}{\left(x^2+y^2\right)^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{x^2+y^2}(2y) = \frac{2y}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{(x^2+y^2)\cdot 2 - 2y\cdot 2y}{(x^2+y^2)} = \frac{2\,(x^2-y^2)}{(x^2+y^2)^2}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Example 6.37

If $u = xy + \sin(xy)$, then show that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$.

Solution:

$$\begin{aligned} u &= xy + \sin(xy) \\ \frac{\partial u}{\partial x} &= y + y\cos(xy) \\ \frac{\partial u}{\partial y} &= x + x\cos(xy) \\ \frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) \\ &= 1 + x\left(-\sin\left(xy\right)\cdot y\right) + \cos\left(xy\right) \\ &= 1 - xy\sin(xy) + \cos(xy) \quad \ldots (1) \\ \frac{\partial^2 u}{\partial y \partial x} &= \frac{\partial}{\partial y}\left(y + y\cos\left(xy\right)\right) \\ &= 1 + \cos\left(xy\right) + y\left(-\sin\left(xy\right)x\right) \\ &= 1 - xy\sin\left(xy\right) + \cos\left(xy\right) \ldots \quad (2) \end{aligned}$$

From (1) and (2), we get

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}.$$

Example 6.38

Verify Euler's theorem for the function $u = \frac{1}{\sqrt{x^2+y^2}}$.

Solution:

$$u(x,y) = \left(x^2+y^2\right)^{-\frac{1}{2}}$$

$$u(tx,ty) = \left(t^2x^2 + t^2y^2\right)^{-\frac{1}{2}} = t^{-1}\left(x^2+y^2\right)^{-\frac{1}{2}}$$

$\therefore$ u is a homogeneous function of degree -1

By Euler's theorem, $x.\frac{\partial u}{\partial x} + y.\frac{\partial u}{\partial y} = (-1)u = -u$

Verification:

$$u = \left(x^2+y^2\right)^{-\frac{1}{2}}$$

$$\frac{\partial u}{\partial x} = -\frac{1}{2}\left(x^2+y^2\right)^{-\frac{3}{2}}\cdot 2x = \frac{-x}{\left(x^2+y^2\right)^{-\frac{3}{2}}}$$

$$x\cdot\frac{\partial u}{\partial x} = \frac{-x^2}{\left(x^2+y^2\right)^{-\frac{3}{2}}}$$

$$\frac{\partial u}{\partial y} = -\frac{1}{2}\left(x^2+y^2\right)^{-\frac{3}{2}}\cdot 2y = \frac{-y}{\left(x^2+y^2\right)^{-\frac{3}{2}}}$$

$$y\cdot\frac{\partial u}{\partial y} = \frac{-y^2}{\left(x^2+y^2\right)^{-\frac{3}{2}}}$$

$$\therefore x\cdot\frac{\partial u}{\partial x} + y\cdot\frac{\partial u}{\partial y} = \frac{-(x^2+y^2)}{(x^2+y^2)^{-\frac{3}{2}}}$$

$$= (-1)\frac{1}{\sqrt{x^2+y^2}} = (-1)u = -u$$

Hence Euler's theorem verified.

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